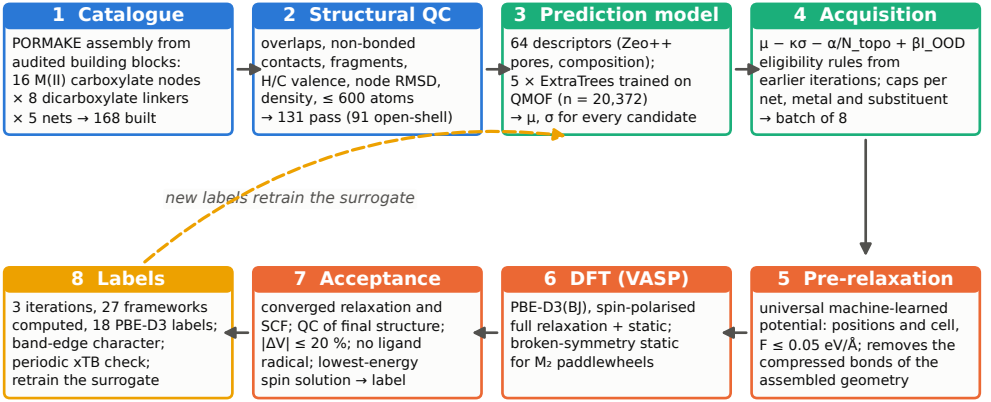
1 Catalogue
PORMAKE assembly from audited building blocks: 16 M(II) carboxylate nodes × 8 dicarboxylate linkers × 5 nets → 168 built
2 Structural QC
overlaps, non-bonded contacts, fragments, H/C valence, node RMSD, density, ≤ 600 atoms → 131 pass (91 open-shell)
3 Prediction model
64 descriptors (Zeo++ pores, composition); 5 × ExtraTrees trained on QMOF (n = 20,372) → μ, σ for every candidate
4 Acquisition
μ − κσ − α/N_topo + βI_OOD eligibility rules from earlier iterations; caps per net, metal and substituent → batch of 8
5 Pre-relaxation
universal machine-learned potential: positions and cell, F ≤ 0.05 eV/Å; removes the compressed bonds of the assembled geometry
6 DFT (VASP)
PBE-D3(BJ), spin-polarised full relaxation + static; broken-symmetry static for M₂ paddlewheels
7 Acceptance
converged relaxation and SCF; QC of final structure; |ΔV| ≤ 20 %; no ligand radical; lowest-energy spin solution → label
8 Labels
3 iterations, 27 frameworks computed, 18 PBE-D3 labels; band-edge character; periodic xTB check; retrain the surrogate
new labels retrain the surrogate